

Mark schemes

Q1.

- (a) $10 \operatorname{cosec}^2 x = 16 - 11 \cos x$
 $10(1 + \cot^2 x) = 16 - 11 \cot x$
 $10 \cot^2 x + 11 \cot x - 6 = 0$

AG Must see evidence of correct identity
 and no errors.

B1

1

- (b) Attempt at factors, giving $\pm 10 \cot^2 x \pm 6$
 when expanded.

Use of formula: condone one error

M1

$$(5 \cot x - 2)(2 \cot x + 3) (= 0)$$

Correct factors

A1

$$\left(\cot x = \frac{2}{5}, -\frac{3}{2} \right)$$

$$\tan x = \frac{5}{2}, -\frac{2}{3}$$

1st A1 must be earned
Condone AWRT – 0.67
ISW if x values attempted

A1,A1

Alternative 1

$$10 \cot^2 x + 11 \cot x - 6 = 0$$

$$10 \frac{\cos^2 x}{\sin^2 x} + 11 \frac{\cos x}{\sin x} - 6 = 0$$

$$10 \cos^2 x + 11 \cos x \sin x - 6 \sin^2 x = 0$$

$$(5 \cos x - 2 \sin x)(2 \cos x + 3 \sin x) (= 0)$$

Attempt at factors, gives
 $\pm 10 \cos^2 x \pm 6 \sin^2 x$ when explained

(M1)

As above

(A1)

$$(5 \cos x = 2 \sin x \quad 2 \cos x = -3 \sin x)$$

$$\frac{5}{2} = \tan x \quad -\frac{2}{3} = \tan x$$

1st A1 must be earned
Condone AWRT –0.67
ISW if x values attempted

(A1),(A1)

Alternative 2

$$10 + 11 \tan x - 6 \tan^2 x = 0$$

$$(5 - 2 \tan x)(2 + 3 \tan x) (= 0)$$

Attempt at factors gives $\pm 10 \pm 6 \tan^2 x$

(M1)(A1)

$$\tan x = \frac{5}{2}, -\frac{2}{3}$$

1st A1 must be earned
Condone AWRT –0.67
ISW if x values attempted

(A1),(A1)

4

[5]

Q2.

(a) $y = e^x \rightarrow e^{2x} - 1$
Stretch (I)

scale factor $\frac{1}{2}$ (II)

I + (II or III)

M1

in x -direction (III)

I + II + III

A1

Translation

Allow “translate”

E1

$$\begin{bmatrix} 0 \\ -1 \end{bmatrix}$$

OE “1 unit down” etc

B1
4

(b) $x = 0$ $y = 6$ or $(0, 6)$

Both coordinates must be stated, not simply 6 marked on diagram

B1
1

(c) (i) $e^{2x} - 1 = 4e^{-2x} + 2$
 $e^{4x} - e^{2x} = 4 + 2e^{2x}$
 or $(e^{2x})^2 - e^{2x} = 4 + 2e^{2x}$

Multiplying both sides by e^{2x}

M1

$$(e^{2x})^2 - 3e^{2x} - 4 = 0$$

AG With no errors seen

A1
2

(ii) $(e^{2x} - 4)(e^{2x} + 1)$
 $(e^{2x} \pm 4)(e^{2x} \pm 1)$

M1

$$x = \ln 2 \text{ or } \frac{1}{2} \ln 4$$

A1

Reject $e^{2x} = -1$ OE

Eg $e^{2x} > 0$, $e^{2x} \neq -1$, impossible etc

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A1
3

(d) $\int (4e^{-2x} + 2) dx$ (I)
 $= \left[\frac{4e^{-2x}}{-2} + 2x \right]_0^{\ln 2}$

I or II attempted and e^{-2x} or e^{2x} integrated correctly

M1

$$= \left(\frac{4e^{-2\ln 2}}{-2} + 2\ln 2 \right) - \left(\frac{4}{-2} + 0 \right)$$

F [‘their $\ln 2$ ’ from (c)(ii)] – F[0]

m1

$$= -\frac{1}{2} + 2 \ln 2 + +2 = \frac{3}{2} + 2 \ln 2$$

$$\int (e^{2x} - 1) dx \quad (II)$$

$$= \left[\frac{e^{2x}}{2} - x \right]_0^{\ln 2}$$

Both I and II correctly integrated

A1

$$= \left(\frac{e^{2 \ln 2}}{2} - \ln 2 \right) - \left(\frac{1}{2} - 0 \right)$$

$$= 2 - \ln 2 - \frac{1}{2} = \frac{3}{2} - \ln 2$$

$$A = \left(\frac{3}{2} + 2 \ln 2 \right) - \left(\frac{3}{2} - \ln 2 \right)$$

Attempt to find difference of 'their I – their II'

B1ft

$$= 3 \ln 2 \text{ or } \ln 8 \text{ or } \frac{3}{2} \ln 4 \quad \text{OE}$$

CSO must be exact

A1

Alternative

$$A = \int (4e^{-2x} + 2) dx - \int (e^{2x} - 1) dx$$

Condone functions reversed

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(B1)

$$= \int_0^{\ln 2} (4e^{-2x} - e^{2x} + 3) dx$$

$$= \left[\frac{4e^{-2x}}{-2} - \frac{e^{2x}}{2} + 3x \right]_0^{\ln 2}$$

e^{2x} or e^{-2x} correctly integrated

(M1)(A1)

$$= \left(-2e^{-2 \ln 2} - \frac{1}{2}e^{2 \ln 2} + 3 \ln 2 \right) - \left(-2 - \frac{1}{2} \right)$$

Correct substitution of their $\ln 2$ from

(c)(ii) into their integrated expression

(m1)

$$= 3 \ln 2 \text{ or } \ln 8 \text{ or } \frac{3}{2} \ln 4 \quad \text{OE}$$

CSO must be exact

(A1)

5

[15]

Q3.

(a) (i) $(x + 1)^2$

$$p = 1$$

B1

$$+ 4$$

$$q = 4$$

B1

2

(ii) $(x + 1)^2 \geq 0 \Rightarrow (x + 1)^2 + 4 > 0$
 $(\Rightarrow x^2 + 2x + 5 > 0 \text{ for all values of } x)$

*Condone if they say $(x + 1)^2$ positive
and adding 4 so always positive*

E1

1

(b) (i) $x = -1$ or $y = 4$

$$\text{ft their } x = -p \text{ or } y = q$$

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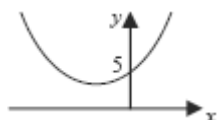
M1

Minimum point is $(-1, 4)$

A1

2

(ii)



Sketch roughly as shown

B1

y-intercept 5 or (0, 5) marked or stated

B1

2

- (c) Translation (not shift, move etc)
and NO other transformation stated

E1

through $\begin{bmatrix} -1 \\ 4 \end{bmatrix}$ (or 1 left, 4 up etc)
either component correct or ft their $-p, q$

M1

correct translation
M1, A1 independent of E mark

A1

3

[10]

Q4.

(a) $3e^x = 4$
 $e^x = \frac{4}{3}$

M1

$x = \ln \frac{4}{3}$

A1

2

(b) (i) $3e^x + 20e^{-x} = 19$
 $3y + \frac{20}{y} = 19$ or $3e^{2x} + 20 = 19e^x$
 $3y^2 - 19y + 20 = 0$
AG

B1

1

(ii) $(3y - 4)(y - 5) = 0$
 $y = \frac{4}{3}, 5$

B1

$$\therefore x = \ln \frac{4}{3}, \ln 5$$

In (their + ve y's)

M1A1

3

[6]

Q5.

(a) (i) $(x - 4)^2$ or $p = 4$

B1

+ 1 or $q = 1$

ISW for $p = -4$ if $(x - 4)^2$ seen

B1

2

(ii) (Minimum value is) 1

Correct or FT "their q " (NOT coords)

B1ft

1

(iii) (Minimum occurs when $x =$) 4

Correct or FT "their p " – may use calculus

*Condone (p , **) for this B1 mark*

B1ft

1

(b) (i) $(x - 5)^2 = x^2 - 10x + 25$

B1

1

(ii) $(x - 5)^2 + (7 - x - 4)^2$

Condone one slip in one bracket

May be seen under $\sqrt{\quad}$ sign

M1

$$= (x - 5)^2 + (3 - x)^2$$

$$= x^2 - 10x + 25 + 9 - 6x + x^2$$

From a fully correct expression

A1

$$AB^2 = 2x^2 - 16x + 34$$

$$= 2(x^2 - 8x + 17)$$

AG CSO

A1

3

(iii) Minimum $AB^2 = 2 \times$ "their (a)(ii)"

Or use of their $x = 4$ in expression

Or use of their $B(4, 3)$ and $A(5, 4)$ in distance formula

M0 if calculus used

Answer only of $2 \times$ "their (a)(ii)" scores

M1, A0

M1

$$\text{Minimum } AB = \sqrt{2}$$

A1

2

[10]

Q6.

(a) $k(x^2 + 3) = 2x + 2$

$$\Rightarrow kx^2 - 2x + 3k - 2 = 0$$

AG OE all terms on one side and $= 0$

B1

1

(b) (i) Discriminant $= (-2)^2 - 4k(3k - 2)$

Condone one slip (including x is one slip)

Condone 2^2 or 4 as first term

M1

$$= 4 - 12k^2 + 8k$$

condone recovery from missing brackets

A1

Two distinct real roots $\Rightarrow b^2 - 4ac > 0$

$$4 - 12k^2 + 8k > 0$$

"their discriminant in terms of k " > 0

Not simply the statement $b^2 - 4ac > 0$

B1ft

$$\Rightarrow 12k^2 - 8k - 4 < 0$$

Change from > 0 to < 0 and divide by 4

$$\Rightarrow 3k^2 - 2k - 1 < 0$$

AG CSO

A1

4

(ii) $(3k + 1)(k - 1)$

Correct factors or correct use of formula

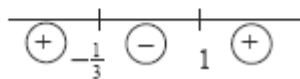
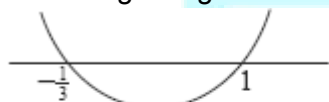
M1

Critical values 1 and $-\frac{1}{3}$

May score M1, A1 for correct critical values seen as part of incorrect final answer with or without working

A1

Use of sign diagram or sketch



If previous A1 earned, sign diagram or sketch must be correct for M1

M1

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Otherwise, M1 may be earned for an attempt at the sketch or sign diagram using their critical values.

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$$\Rightarrow -\frac{1}{3} < k < 1 \quad \text{or} \quad 1 > k > -\frac{1}{3}$$

*Full marks for correct final answer with or without working
≤ loses final A mark*

A1

condone $-\frac{1}{3} < k$ AND $k < 1$ for full marks but not OR or “,” instead of AND

Answer only of $1 < k < -\frac{1}{3}$ or

$k < -\frac{1}{3}; k < 1$ etc scores M1, A1, M0 since
 the correct critical values are evident

Answer only of $\frac{1}{3} < k < 1$ etc where
 critical values are not both correct gets M0, M0

4

[9]

Q7.

(a) $2 \ln x = 5$

$$\ln x = \frac{5}{2} \quad x = e^{\frac{5}{2}}$$

B1

1

(b) $2 \ln x + \frac{15}{\ln x} = 11$

$$2(\ln x)^2 - 11 \ln x + 15 = 0$$

Forming quadratic equation in $\ln x$
 condone poor notation

M1

$$(2 \ln x - 5)(\ln x - 3) = 0$$

Attempt at factorisation/formula

m1

$$\ln x = \frac{5}{2}, 3 \quad \text{condone } 2 \ln x = 5$$

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A1

$$x = e^{\frac{5}{2}}, e^3$$

[SC for substituting $x = e^{\frac{5}{2}}$ or equivalent
 into equation and verifying B1 $\left(\frac{1}{5}\right)$]

A1, A1

5

[6]

Q8.

(a) $x^2 + (y - 5)^2$
 $b = 5$

B1

RHS = 5
 $k = 5$

B1

2

(b) (i) Centre (0, 5)
FT their b from part (a)

B1ft

1

(ii) Radius = $\sqrt{5}$
FT their k from part (a); RHS must be > 0

B1ft

1

(c) (i) $x^2 + 4x^2 - 20x + 20 = 0$
May substitute into original or "their (a)"

M1

$\Rightarrow x^2 - 4x + 4 = 0$
 CSO; AG

A1

2

(ii) $(x - 2)^2 = 0$ or $x = 2$

M1

Repeated root implies tangent
Or $b^2 - 4ac$ shown = 0 plus statement

E1

Point of contact is $P(2, 4)$

A1

3

(d) $(CQ^2 =) 1^2 + 1^2$
FT their C

M1

$$\sqrt{2} < \sqrt{5} \Rightarrow Q \text{ lies inside circle}$$

CQ or CQ^2 OE must appear for A1

A1CSO

2

[11]

Q9.

(a) $(9 + x)(1 - x)$

$\pm(9 \pm x)(1 \pm x)$

M1

Correct factors

A1

2

(b) $25 - (x^2 + 8x + 16) = 9 - 8x - x^2$

AG

B1

1

(c) (i) $x = -4$ is line of symmetry

B1

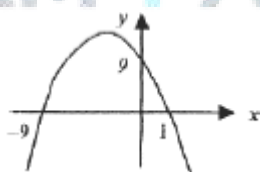
1

(ii) Vertex is $(-4, 25)$

B1, B1

2

(iii)



General $\cap$ shape

M1

-9 and 1 marked on x -axis or stated

B1

9 marked on y -axis and maximum to the left of y -axis

Must continue below x -axis at both ends

A1

3

Q10.

(a) $x^2 + 7 = k(3x + 1) \Rightarrow x^2 - 3kx + 7 - k = 0$

AG

B1

1

(b) $b^2 - 4ac = (-3k)^2 - 4(7 - k)$

Clear attempt at $b^2 - 4ac$

Condone slip in one term of expression

M1

(2 distinct roots when) $b^2 - 4ac > 0$

Must involve k

B1

$9k^2 + 4k - 28 > 0$

CSO; AG

A1

3

(c) $(9k - 14)(k + 2)$

Factors or formula correct unsimplified

M1

Critical points -2 and $\frac{14}{9}$

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A1

Sketch  or sign diagram correct

+ve	-ve	+ve
	-2	$\frac{14}{9}$

M1

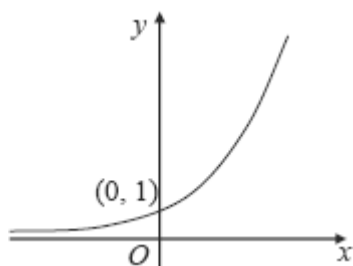
$k < -2, k > \frac{14}{9}$

A1

4

Q11.

(a)



Shape (graph must clearly go below the intersection pt.). Condone if x -axis is a tangent

B1

Only intersection with y -axis at $(0, 1)$ stated/indicated ... (accept 1 on y -axis as equivalent) 0

B1

2

- (b) (i) Stretch **(I)** in x -direction **(II)** scale factor 0.5 **(III)**
*Need **(I)** & one of **(II)**, **(III)***

M1

M0 if >1 transformation

A1

2

- (ii) Translation;

Must be 'Translation' or 'translate (d)' for 1st B mark

B1; *©2025 Exam Papers Practice. All rights reserved.*

$$\begin{bmatrix} -1 \\ 0 \end{bmatrix}$$

*Accept **full** equivalent to vector in words provided linked to 'translation/ move/shift' and **negative** x -direction (Note: B0 B1 is possible)*

B1

ALTn: Stretch **(I)** in y -direction **(II)** scale factor 3 **(III)**
[Mark the alternative as in (b)(i).]

2

(c) (i) $9^x = (3^2)^x = 3^{2x} = (3^x)^2 = Y^2$;

Justifying either $9^x = Y^2$ or $3^{x+1} = 3Y$

M1

$$3^{x+1} = 3^x \times 3^1 = 3Y$$

$$9^x - 3^{x+1} + 2 = 0 \Rightarrow Y^2 - 3Y + 2 = 0$$

$$\Rightarrow (Y-1)(Y-2) = 0$$

AG

A1

2

(ii) $Y = 1 \Rightarrow 3^x = 1 \Rightarrow x = 0$

AG (Accept direct substitution if convinced)

B1

$$Y = 2 \Rightarrow 3^x = 2$$

$$\log_{10} 3^x = \log_{10} 2$$

*Takes logs of both, PI by 'correct' value(s) later.
or $x = \log_3 2$ seen*

M1

$$x \log_{10} 3 = \log_{10} 2$$

Use of $\log 3^x = x \log 3$ or

$$\log_3 2 = \frac{\lg 2}{\lg 3} \quad \text{OE (PI by } \log_3 2 = 0.630 \text{ or } 0.631 \text{ or better)}$$

m1

$$x = \frac{\lg 2}{\lg 3} = 0.630929 \dots = 0.6309 \text{ to 4 dp}$$

*Must show that logarithms have been used
otherwise 0/3*

A1

4

[12]

Q12.

- (a) *L*: straight line with positive gradient and negative intercept on *y*-axis

Line must cross both axes but need not reach the curve

B1

cutting at $(\frac{1}{3}, 0)$ and $(0, -1)$
 (intercepts stated or marked on sketch)

Condone 0.33 or better for $\frac{1}{3}$

B1

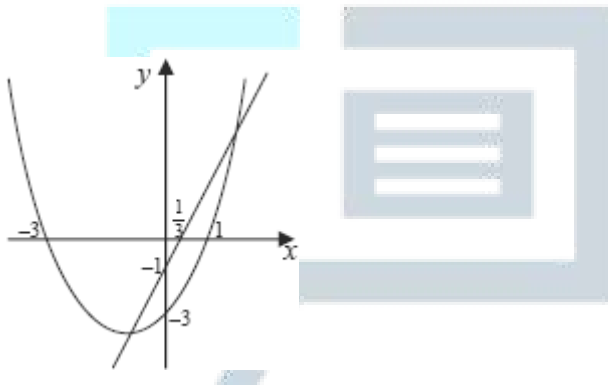
C: attempt at parabola $\cup$ or $\cap$
 through $(-3, 0)$ and $(1, 0)$ or
 values -3 and 1 stated as intercepts
 on x -axis

B1

$\cup$ shaped graph – vertex below x -axis and cutting x -axis twice

M1

through $(0, -3)$ and minimum point to left of y -axis



A1

(y-intercept or coordinates marked)

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5

(b) $(x + 3)(x - 1) = 3x - 1$

M1

$$x^2 + 3x - x - 3 - 3x + 1 = 0$$

$$\Rightarrow x^2 - x - 2 = 0$$

AG; must have “= 0” and no errors

A1

2

(c) $(x - 2)(x + 1) = 0$

$(x \pm 1) (x \pm 2)$ or use of formula (one slip)

M1

$$\Rightarrow x = 2, -1$$

correct values imply M1A1

A1

Substitute one value of x to find y

m1

Points of intersection $(2, 5)$ and $(-1, -4)$

May say $x = 2, y = 5$ etc

SC: $(2, 5) \Rightarrow B2$

$(-1, -4) \Rightarrow B2$ without working

A1

4

[11]

Q13.

(a) $\left(x - \frac{3}{2}\right)^2$

Must have $()^2$ $p = 1.5$

B1

$$+ \frac{7}{4}$$

$$q = 1.75$$

B1

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(b) Minimum value is $\frac{7}{4}$

ft their q or correct value

B1ft

1

(c) **Translation** (and **no other** transformation stated)

*(**not** shift, move, transformation etc)*

E1

through $\begin{bmatrix} 3 \\ 2 \\ 7 \\ 4 \end{bmatrix}$ (or equivalent in words)

*M1 for one component correct
 or ft their p or q values*

M1

CSO; condone 1.5 right and 1.75 up etc

A1

3

[6]

Q14.

(a) $b^2 - 4ac = 16k^2 - 36(k + 1)$
Condone one slip

M1

Real roots: discriminant ≥ 0

B1

$\Rightarrow 16k^2 - 36k - 36 \geq 0$
 $\Rightarrow 4k^2 - 9k - 9 \geq 0$
AG (watch signs)

A1

3

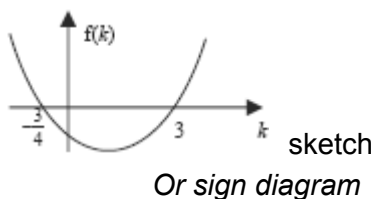
(b) $(4k + 3)(k - 3)$
Or correct use of formula (unsimplified)

M1

critical points $(k =) -\frac{3}{4}, 3$

*Not in a form involving surds
 Values may be seen in inequalities etc*

A1



M1

$k \geq 3, k \leq -\frac{3}{4}$
NMS full marks

Condone use of word "and" but final

answer in a form such as $3 \leq k \leq -\frac{3}{4}$
 scores A0

A1

4

[7]

Q15.

(a) (i) $y = -\frac{3}{5}x + \dots$; Gradient $AB = -\frac{3}{5}$

Attempt to find $y =$ or $\Delta y / \Delta x$ or $\frac{3}{5}$ or $3x/5$

M1

Gradient correct – condone slip in $y = \dots$

A1

2

(ii) $m_1 m_2 = -1$

Stated or used correctly

M1

Gradient of perpendicular = $\frac{5}{3}$
ft gradient of AB

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A1 ✓

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$$\Rightarrow y + 2 = \frac{5}{3}(x - 6)$$

CSO Any correct form eg $y = \frac{5}{3}x - 12$,
 $5x - 3y = 36$ etc

A1

3

(b) Eliminating x or y (unsimplified)

Must use $3x + 5y = 8$; $2x + 3y = 3$

M1

$$x = -9$$

A1

$$y = 7$$

$$B(-9, 7)$$

A1

3

(c) $4^2 + (k + 2)^2 (=25)$ or $16 + d^2 = 25$

Diagram with 3, 4, 5 triangle

M1

$$k = 1$$

A1

Condone slip in one term (or $k + 2 = 3$)

or $k = -5$

SC1 with no working for spotting one correct value of k . Full marks if both values spotted with no contradictory work

A1

3

[11]

Q16.

(a) $b^2 - 4ac = 144 - 4(k + 1)(k - 4)$

Clear attempt at $b^2 - 4ac$

Condone slip in one term of expression

M1

Real roots when $b^2 - 4ac \geq 0$

Not just a statement, must involve k

B1

$$36 - (k^2 - 3k - 4) \geq 0$$

$$\Rightarrow k^2 - 3k - 40 \leq 0$$

AG (watch signs carefully)

A1

3

(b) $(k - 8)(k + 5)$

Factors attempt or formula

M1

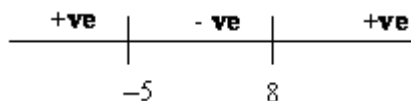
Critical points 8 and -5

A1

Sketch or sign diagram **correct**, must have 8 and -5

M1

$$-5 \leq k \leq 8$$



A1

4

A0 for $-5 < k < 8$ or two separate inequalities unless word AND used

[7]

Q17.

(a) $\cos 2x = 1 - 2\sin^2 x$

B1

1

(b) (i) $3\sin x - \cos 2x = 3\sin x - (1 - 2\sin^2 x)$

Candidate's $\cos 2x$ or $\sin^2 x$

M1

$$= 3\sin x - 1 + 2\sin^2 x$$

AG

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A1

2

(ii) $2\sin^2 x + 3\sin x - 2 = 0$

Soluble quadratic form

M1

$$(2\sin x - 1)(\sin x + 2) = 0$$

Attempt to solve (allow one error in formula, allow sign errors)

M1

$$\sin x = \frac{1}{2} \quad x = 30 \quad x = 150$$

M1

$\sin^{-1}$ and two solutions ($0^\circ < x < 360^\circ$)
A0 if radians

A1

4

Allow misread for $2\sin^2 x + 3\sin x - 1 = 0$
Soluble quadratic form

(M1)

$$\sin x = \frac{-3 \pm \sqrt{17}}{4}$$

Use of formula (allow one error)

(M1)

$x = 16.3^\circ, 163.7^\circ$
Max 3/4

(A1)

(c) $\int \frac{1}{2}(1 - \cos 2x) = \frac{x}{2} - \frac{\sin 2x}{4} (+c)$

M1 – solve integral, must have 2 terms
for $\sin^2 x$ from (a)

M1A1

2

[9]

Q18.

(a) (i) $(x + 5)^2$
 $p = 5$

B1

–6

$q = -6$

B1

2

(ii) $x_{\text{vertex}} = -5$ (or their $-p$)
may differentiate but must have $x = -5$

B1 ✓

$y_{\text{vertex}} = -6$ (or their q)
and $y = -6$. Vertex $(-5, -6)$

B1 ✓

2

(iii) $x = -5$

B1

1

(iv) Translation (not shift, move etc)
and NO other transformation stated

E1

through $\begin{bmatrix} -5 \\ -6 \end{bmatrix}$ (or 5 left, 6 down)
either component correct

M1

M1, A1 independent of E mark

A1

3

(b) $x + 11 = x^2 + 10x + 19$
quadratic with all terms on one side of equation

$\Rightarrow x^2 + 9x + 8 = 0$ or $y^2 - 13y + 30 = 0$

M1

$(x + 8)(x + 1) = 0$ or $(y - 3)(y - 10) = 0$

attempt at formula (1 slip) or to factorise

ml

$\left. \begin{matrix} x = -1 \\ y = 10 \end{matrix} \right\}$ or $\left. \begin{matrix} x = -8 \\ y = 3 \end{matrix} \right\}$

both x values correct

A1

both y values correct and linked

SC $(-1, 10)$ B2, $(-8, 3)$ B2 no working

A1

4

[12]

Q19.

- (a) (i) $f(1) = 1 + 4 - 5$
must find $f(1)$ NOT long division

M1

$\Rightarrow f(1) = 0 \Rightarrow (x - 1)$ is factor
shown = 0 plus a statement

A1

2

- (ii) Attempt at $x^2 + x + 5$
*long division leading to $x^2 \pm x \dots$ or
 equating coefficients*

M1

$f(x) = (x - 1)(x^2 + x + 5)$
 $p = 1, q = 5$ by inspection scores B1, B1

A1

2

- (iii) $(x =)1$ is real root

B1

Consider $b^2 - 4ac$ for their $x^2 + x + 5$
not the cubic!

M1

$$b^2 - 4ac = 1^2 - 4 \times 5 = -19 < 0$$

Hence no real roots (or only real root is 1)
CSO; all values correct plus a statement

A1

3

- (b) (i) $\int \dots dx = \frac{x^2}{4} + 2x^2 - 5x (+c)$
one term correct unsimplified

M1

second term correct unsimplified

A1

all correct unsimplified

A1

3

(ii) $[4+8-10]-\left[\frac{1}{4}+2-5\right]$

*correct use of limits 1 and 2;
F(2) – F(1) attempted*

M1

$$= 4\frac{3}{4}$$

A1

Area of $\Delta = \frac{1}{2} \times 11 = 5\frac{1}{2}$

correct unsimplified

B1

$$\Rightarrow \text{shaded area} = 5\frac{1}{2} - 4\frac{3}{4}$$

*combined integral of $7x - 6 - x^3$ scores
M1 for limits correctly used then*

$$= \frac{3}{4}$$

A3 correct answer with all working correct

A1

EXAM PAPERS PRACTICE 4 [14]

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Q20.

(a) $b^2 - 4ac = 4 - 4(k-1)(2k-3)$

(or seen in formula) condone one slip

M1

Real roots when $b^2 - 4ac \geq 0$

must involve $f(k) \geq 0$ (usually M1 must be earned)

E1

$$4 - 4(2k^2 - 5k + 3) \geq 0$$

$$\Rightarrow -2k^2 + 5k - 3 + 1 \geq 0$$

at least one step of working justifying ≤ 0

$$\Rightarrow 2k^2 - 5k + 2 \leq 0$$

AG

A1

3

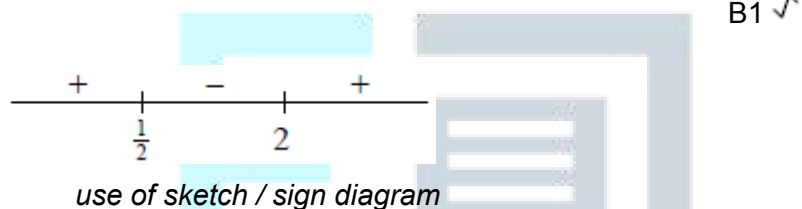
(b) (i) $(2k-1)(k-2)$

B1

1

(ii) (Critical values) $\frac{1}{2}$ and 2

ft their factors or correct values seen on diagram, sketch or inequality or stated



B1 ✓

M1

$$\Rightarrow 0.5 \leq k \leq 2$$

M1A0 for $0.5 < k < 2$ or $k \geq 0.5, k \leq 2$

EXAM PAPERS PRACTICE

A1

3

[7]

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Q21.

(a) (i) $(x-2)^2$

$$p = 2$$

B1

$$+ 5$$

$$q = 5$$

B1

2

(ii) Minimum point (2, 5) or $x = 2, y = 5$

B1 for each coordinate correct or ft

Alt method M1, A1 sketch, differentiation

B2 ✓

2

(b) (i) $12 - 2x = x^2 - 4x + 9$

Or $x^2 - 4x + 9 + 2x = 12$

$\Rightarrow x^2 - 2x - 3 = 0$

AG (be convinced) (must have = 0)

B1

1

(ii) $(x - 3)(x + 1) = 0$

Attempt at factors or quadratic formula or one value spotted

M1

$x = 3, -1$

Both values correct & simplified

A1

Substitute one value of x to find y

May substitute into equation for L or C

M1

Points are (3, 6) and (-1, 14)

y -coordinates correct linked to x values

A1

4

EXAM PAPERS PRACTICE

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[9]

Q22.

(a) $(m + 4)^2 = m^2 + 8m + 16$

Condone $4m + 4m$

B1

$b^2 - 4ac = (m + 4)^2 - 4(4m + 1) = 0$

*$b^2 - 4ac$ (attempted and involving m 's and no x 's)
or $b^2 - 4ac = 0$ stated*

M1

$m^2 + 8m + 16 - 16m - 4 = 0$

$$\Rightarrow m^2 - 8m + 12 = 0$$

*AG (be convinced – all working correct –
= 0 appearing more than right at the end)*

A1

3

(b) $(m - 2)(m - 6) = 0$

Attempt at factors or quadratic formula

M1

$$m = 2, m = 6$$

SC B1 for 2 or 6 only without working

A1

2

[5]

Q23.

(a) $2\log_a n - \log_a (5n - 24) = \log_a 4$

$$\Rightarrow \log_a n^2 - \log_a (5n - 24) = \log_a 4$$

A law of logs used

M1

$$\Rightarrow \log_a \left[\frac{n^2}{5n - 24} \right] = \log_a 4$$

*A second law of logs used leading to both
sides being single log terms or single log
term on LHS with RHS = 0*

M1

$$\Rightarrow \frac{n^2}{5n - 24} = 4$$

$$\Rightarrow n^2 - 20n + 96 = 0$$

CSO. AG

A1

3

(b) $\Rightarrow (n - 8)(n - 12) = 0$

Accept alternatives eg formula, completing of sq..

M1

$$\Rightarrow n = 8, 12$$

A1

2

[5]

Q24.

(a) $(x + 4)^2$

$p = 4$

B1

+3

$q = 3$

B1

2

(b) $(x + 4)^2 = -3$ or "their" $(x + p)^2 = -q$
 Or discriminant = $64 - 76$

M1

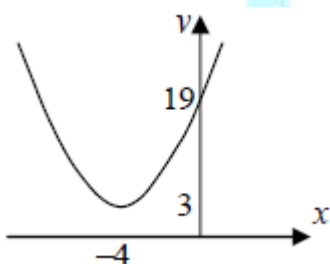
No real square root of -3

Disc < 0 so no real roots (all correct figs)

A1

2

(c)



Minimum $(-4, 3)$

ft their $-p$ and q (or correct)

B1 ✓

graph

Parabola (vertex roughly as shown)

B1

Crossing at $y = 19$ marked or $(0, 19)$ stated

B1

3

(d) Translation (and no additional transformation)

Not shift, move, transformation, etc

E1

through $\begin{bmatrix} -4 \\ 3 \end{bmatrix}$

One component correct eg 3 units up

M1

*All correct – if not vector – must say 4 units
in negative x-direction, to left etc*

A1

3

[10]

Q25.

(a) (i) $(x - 2)^2$

Attempt to complete square for x

M1

centre has x -coordinate = 2

M1 implied if value correct or -2

A1

and y -coordinate = 0

Centre (2,0)

B1

EXAM PAPERS PRACTICE

3

(ii) RHS = 18

Withhold if circle equation RHS incorrect

B1

Radius = $\sqrt{18}$

Square root of RHS of equation (if > 0)

M1

Radius = $3\sqrt{2}$

A1

3

- (b) Perpendicular bisects chord so need to use
Length of 4

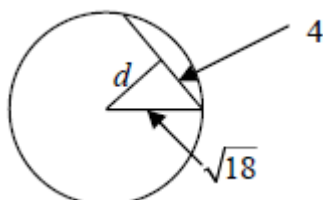
B1

$$d^2 = (\text{radius})^2 - 4^2$$

M1

$$d^2 = 18 - 16 \text{ so perpendicular distance} = \sqrt{2}$$

A1



3

(c) (i) $x^2 + (2k - x)^2 - 4x - 14 = 0$

M1

$$(2k - x)^2 = 4k^2 - 4kx + x^2$$

B1

$$\Rightarrow 2x^2 + 4k^2 - 4kx - 4x - 14 = 0$$

$$(\Rightarrow x^2 + 2k^2 - 2kx - 2x - 7 = 0)$$

$$\Rightarrow x^2 - 2(k+1)x + 2k^2 - 7 = 0$$

AG (be convinced about algebra and = 0)

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A1

3

(ii) $4(k+1)^2 - 4(2k^2 - 7)$ Papers Practice. All rights reserved.

$"b^2 - 4ac"$ in terms of k (either term correct)

M1

$$4k^2 - 8k - 32 = 0 \text{ or } k^2 - 2k - 8 = 0$$

$b^2 - 4ac = 0$ correct quadratic equation in k

A1

$$(k-4)(k+2) = 0$$

Attempt to factorise, solve equation

m1

$$k = -2, k = 4$$

SC B1, B1 for -2, 4 (if M0 scored)

A1

4

(iii) Line is a tangent to the circle

Line touches circle at one point etc

E1

1

[17]

Q26.

(a) $k^2 + 10k + 25 - 12k^2 - 24k$

Condone one slip

M1

$$= -11k^2 - 14k + 25$$

No ISW here

A1

2

(b) (i) Real roots when " $b^2 - 4ac$ " ≥ 0

Non-negative discriminant (stated / used)

B1

$$(k + 5)^2 - 12k(k + 2)$$

*Finding $b^2 - 4ac$ in terms of k
Or factorisation attempt.*

M1

$(k - 1)(11k + 25)$ attempted to be shown

m1

equal to $11k^2 + 14k - 25$

$$-11k^2 - 14k + 25 \geq 0$$

$$\Rightarrow (k - 1)(11k + 25) \leq 0$$

Real roots condition correct and ...

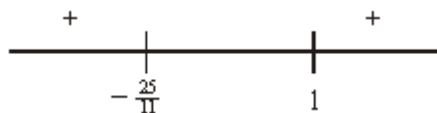
AG (be convinced about inequality)

A1

5

(ii) (Critical values) 1 and $-\frac{25}{11}$ seen

B1



Sketch or sign diagram

M1

$$\Rightarrow -\frac{25}{11} \leq k \leq 1$$

A1

3

[10]

Q27.

(a) $(x - 3)^2$

$p = 3$
 $+ 7$
 $q = 7$

B1

B1

2

(b) (i) Vertex (3, 7)
ft their p

EXAM PAPERS PRACTICE

ft their q

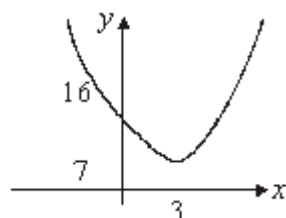
B1ft

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B1ft

2

(ii) graph



parabola (ft on vertex approx position)
correct with y = 16 marked or stated

M1A1

2

- (iii) Line of symmetry $x = 3$
Must have correct equation

B1

1

- (c) Translation (and no additional transf'n) through $\begin{bmatrix} 3 \\ 7 \end{bmatrix}$
Not shift, move, transformation, etc

E1

one part correct eg 7 units up

M1

all correct – if not vector – must say 3 units in positive x-direction etc

A1

3

[10]

Q28.

- (a) $p(x) = x^3 + x^2 + 3x - 2x^2 - 2x - 6$
Condone one slip

M1

$$= x^3 - x^2 + x - 6$$

$$a = -1, b = 1$$

A1

2

- (b) Considering $x^2 + x + 3 = 0$ and attempting to solve or use discriminant

$$b^2 - 4ac = 1 - 12 = -11$$

M1

$$b^2 - 4ac < 0 \Rightarrow \text{no real roots}$$

CSO

A1

Only real root is 2

$$x = 2$$

B1

3

[5]

Q29.

(a) $mx - 1 = x^2 - 5x + 3$

Strict mark here – no trailing equals signs

$$\Rightarrow x^2 - 5x - mx + 4 = 0$$

$$\Rightarrow x^2 - (5 + m)x + 4 = 0$$

AG

(be convinced about algebra and = 0)

B1

1

(b) $(5 + m)^2 - 16 = 0$

Equal roots when “ $b^2 - 4ac$ ” = 0 used

M1

$$5 + m = (\pm)4 \quad \text{or} \quad (m + 1)(m + 9) = 0$$

Square root or factor/formula attempt

m1

$$m = -1$$

A1

$$m = -9$$

SC B1, B1 only for -1, -9 (no working)

A1

4

(c) Line is a tangent to the curve

Line touches curve, cuts at one point etc

E1

1

[6]

Q30.

(a) (i) $4 \frac{\sin \theta}{\cos \theta} \sin \theta = 15$

$$\Rightarrow 4 \sin^2 \theta = 15 \cos \theta$$

AG Be convinced

B1

1

(ii) $\sin^2 \theta + \cos^2 \theta = 1$

OE seen

M1

$$4(1 - \cos^2 \theta) = 15 \cos \theta$$

$$4 \cos^2 \theta + 15 \cos \theta - 4 = 0$$

AG Be convinced

A1

2

(b) (i) $(4c - 1)(c + 4) = 0$

Factorisation or formula or completion of square

M1

$$c = -4, c = \frac{1}{4}$$

Both values

A1

2

(ii) Since $-1 \leq \cos \theta \leq 1$, the only possible value for $\cos \theta$ is $\frac{1}{4}$

AG convincingly explained
(Condone strict inequalities)

ft provided candidates answers for c

*are $\frac{1}{4}$ and a value
k such that $k > 1$ or $k < -1$*

E1ft

1

(iii) $\theta = 75.5^\circ$

B1

$$\theta = 284.5^\circ$$

*Ft on $[360 - c's 75.5^\circ]$ as only other
solution in the given interval*

B1ft

2

(c) $\Rightarrow \cos 4x = \frac{1}{4}$

Links with previous parts. P1

M1

$$x = 19^\circ, 71^\circ$$

ft on (iii)/4....(only ft if 2 answers in given range).

A1ft

2

[10]

Q31.

(a) $x = ut \cos \alpha$

M1

$$t = \frac{x}{u \cos \alpha}$$

A1

$$y = -\frac{1}{2}gt^2 + ut \sin \alpha$$

Must have correct signs

M1

$$y = -\frac{1}{2}g\left(\frac{x}{u \cos \alpha}\right)^2 + u\left(\frac{x}{u \cos \alpha}\right) \sin \alpha$$

M1

$$y = -\frac{gx^2}{2u^2 \cos^2 \alpha} + \frac{x \sin \alpha}{\cos \alpha}$$

$$y = -\frac{gx^2}{2u^2} (1 + \tan^2 \alpha) + x \tan \alpha$$

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A1

$$k = -\frac{10(2k)^2}{2u^2} (1 + \tan^2 \alpha) + 2k \tan \alpha$$

M1

$$u^2 = -20k (1 + \tan^2 \alpha) + 2u^2 \tan \alpha$$

$$20k \tan^2 \alpha - 2u^2 \tan \alpha + u^2 + 20k = 0$$

AG

A1

7

(b) Pass through $P \Rightarrow$ Discriminant ≥ 0

$$(-2u^2)^2 - 4(20k)(u^2 + 20k) \geq 0$$

OE must be seen

M1A1

$$4u^4 - 80ku^2 - 1600k^2 \geq 0$$

$$u^4 - 20ku^2 - 400k^2 \geq 0$$

AG

A1

3

[10]

Q32.

(a) $y = ut \sin \alpha - \frac{1}{2}gt^2$

M1A1

$$x = ut \cos \alpha$$

M1

$$t = \frac{x}{u \cos \alpha}$$

A1

$$y = u \left(\frac{x}{u \cos \alpha} \right) \sin \alpha - \frac{1}{2}g \left(\frac{x}{u \cos \alpha} \right)^2$$

M1

$$y = x \tan \alpha - \frac{gx^2}{u^2 \cos^2 \alpha}$$

$$y = x \tan \alpha - \frac{gx^2}{2u^2} (1 + \tan^2 \alpha)$$

Answer given

A1

6

(b) (i) $1 = R \tan \alpha - \frac{10R^2}{2u^2} (1 + \tan^2 \alpha)$

M1

$$5R^2 \tan^2 \alpha - u^2 R \tan \alpha + 5R^2 + u^2 = 0$$

Answer given

A1

2

(ii) For real solutions of the quadratic:

$$u^4 R^2 - 20R^2(5R^2 + u^2) \geq 0$$

M1

$$R^2 \leq \frac{u^4 - 20u^2}{100}$$

$$R^2 \leq \frac{u^2(u^2 - 20)}{100}$$

Answer given

A1

2

(iii)

$$5^2 \leq \frac{u^2(u^2 - 20)}{100}$$

$$u^4 - 20u^2 - 2500 \geq 0$$

Condone equation

M1

$$u_{\min}^2 = 61.0 \quad (\text{or } 10 + \sqrt{2600})$$

A1

$$u_{\min} = 7.81$$

3 sf required

A1F

3

[13]